# Boss Blimp

### Intro

The boss blimp is an encounter that requires the player to utilize their throwing, movement and grab controls. This boss is intended to mimic the behavior of the regular Blimp enemy, though with enhanced gun defenses and a modified attack pattern.

The Boss Blimp moves into the vicinity of the player and make an initial bombing run on them. After bombing the player, the blimp will use a figure-eight attack the player and drop bombs in the crossover. If the Blimp takes damage or loses its shield it will retreat to the 'Recovery' pattern.

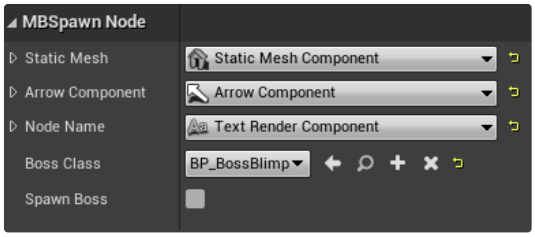
The boss blimp is not grabbable and is too heavy for damage by slapping.

### Type

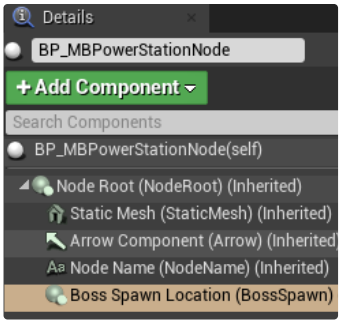
Airborne Blimp, Bomber.

### Blimp Spawn

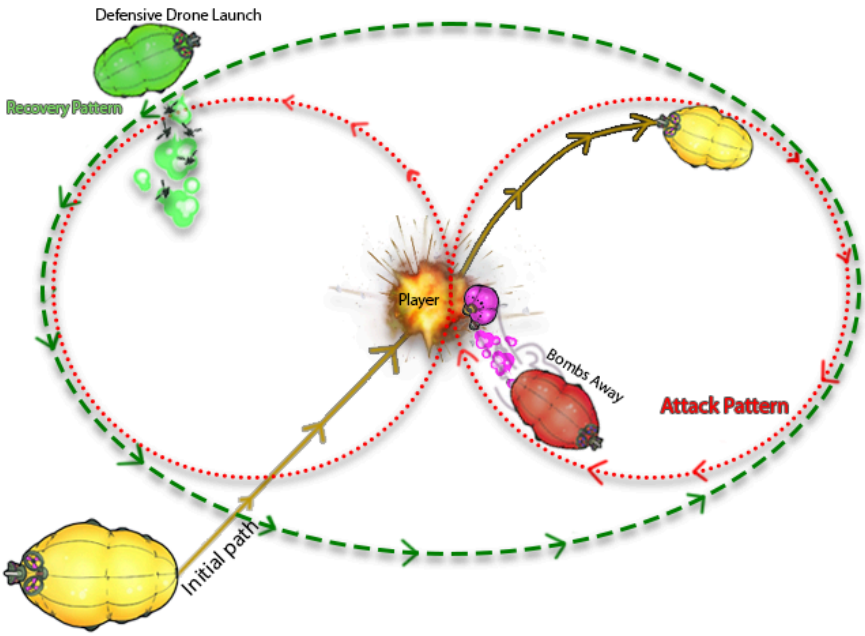
The Boss Blimp Spawns when key infrastructure is destroyed by the player.
The Designer may choose which structures spawn a blimp boss by checking the 'Spawn Boss' box on the corresponding generation node.
Bosses can be spawned from Shield Generators and Power Plants on destruction and Power Stations.




The Boss Blimp will spawn at a  location placed by the Level Designer.




**Movement**




The blimp has two patterns.

In it's main 'Attack' posture it moves in a figure-eight over the player, dropping bombs as it passes over the player.

In it's defensive posture it moves faster and uses gunfire (bullets) to deter the player.

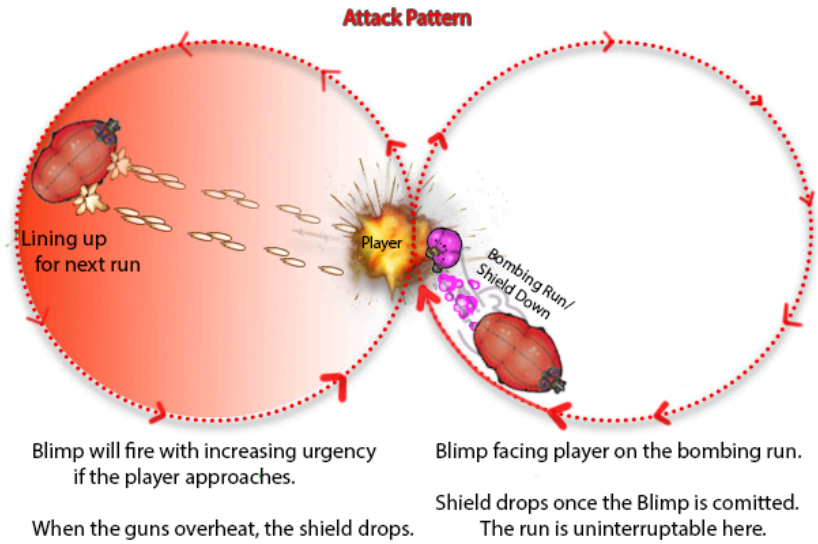
| Variable | Explanation | Location |
|---|---|---|
| **Ideal Height** | The ideal height above ground level at which the boss attempts to fly. | Blimp Bp |

| | | |
|---|---|---|
| **Base Acceleration** | The rate at which the blimp accelerates. | blimp BP |
| **Acceleration Rate** | How quickly the blimp accellerates, multiplier. | blimp BP |
| **Cruising Acceleration** | The speed at which the boss travel when they are not fleeing. | blimp BP |
| **Top Acceleration** | The rate at which blimps move when inside the attack range and when fleeing. | blimp BP |
| **Maximum Firing Range** | The upward limit of the blimps gun range | blimp BP |
| **Threatened Distance** | The distance at which the blimp begins to accelerate and shoot at the player. | blimp BP |
| **Punch Damage Threshold** | How hard the boss needs to be punched to take damage. | blimp BP |
| **Bombing Run Accelleration** | Affects the rate of acceleration when the boss is on the bombing run. | blimp BP |
| **DeflectionPower** | How much energy to add to deflected objects that hit the shield | blimp BP |

#### Rotation

The blimp will be rotated towards it's current heading and the next node on it's path.
It only faces the player on the bombing run. When the blimp is in it's defensive pattern it will attempt to point it's guns (broadside) at the player.

#### Attack




The blimp will attack the player in a figure-eight pattern.
The bombing run is signalled by the sound of the bomb-launcher being armed and the appearance of a bomb mounted below the blimp. The bomb has a timed fuse indicated by a blinking light.

As the blimp aligns with the player and initiates the bombing run, it accelerates and drops the shield to loose the bomb. The bomb drops toward the player, bouncing if it collides with the ground. As the fuse nears it's end the flashes quicken. The player can pick this bomb up and throw it with the timer intact, or crush it to stop the timer.

After the bombing run, the blimp raises it's shields as it hits the outer edge of it's path. If the player advances on the Blimp at this stage it will begin defensively shooting bullets at the player. The guns will fire with increasing haste as the player approaches and they will begin to glow.
If the guns overheat the shield will drop until the bombing run is ended.

| Variable | Explanation | |
|---|---|---|
| **Shield Drop Distance** | The distance from the center of the figure-eight attack pattern the boss must be for the shield to drop during the bombing run. | |
| **Number of Laser Shots Per Burst** | The initial number of shots in the laser burst. | |
| **Starting Burst Cooldown Time** | The time between bursts fired. | |
| **Base Gun Fire Rate** | Dictates the RPM of the laser gun before it begins to overheat | |
| **Gun Heat Per Shot** | Tracks how much heat the gun takes on each shot. | |
| **Current Gun Heat** | Tracks the current heat of the gun - should be left at 0. | |
| **Gun Cooldown Value** | How many points the heat will decrease by every Cooldown Rate cycle. | |
| **Gun Cooldown Rate** | Dictates how often the gun will cool by the cooldown in seconds. | |
| **Number of Bombs to Drop** | Dictates how many bombs the boss will | |

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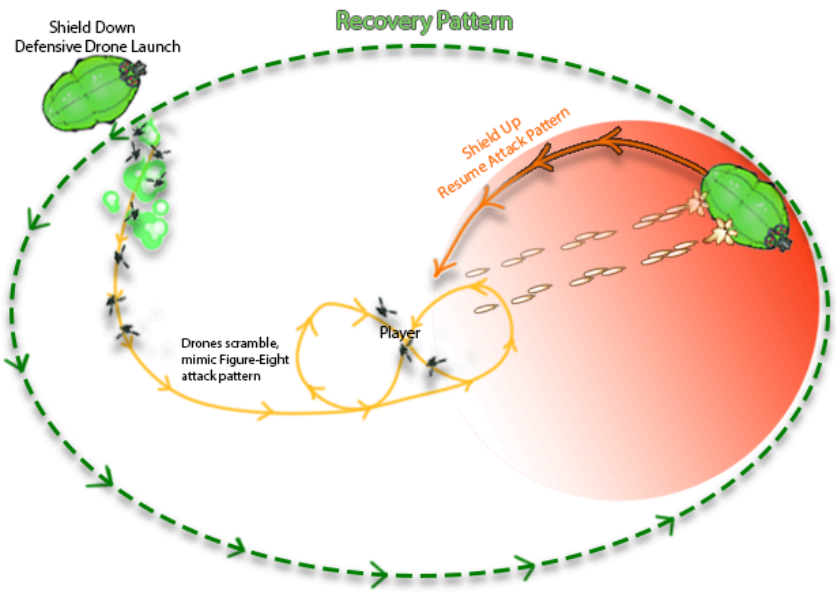
|---|---|---|
| | drop in it's undamaged state. | |
| **Bomb Increase Per Hit** | Dictates the increase in how many bombs the boss will drop each time the boss takes damage. | |
| **Bomb Drop Interval** | The amount of time between bombs dropped on a run. | |

**Defence**




Blimp maintains defensive posture for one rotation of the player after taking damage.

**Ignore 'Drone launch'**

The Blimp is surrounded by an energy shield.

If the player touches the energy shield, the player takes damage and will be electrocuted (😫 Status Effects) making the hand numb and useless so the player has to shake it to recover control of that hand quickly (else the effects goes away after 8 seconds on its own, shaking it reduces the time to just 2 seconds). Upon touch the boss will act according to it's retreat sequence.

The shield is invulnerable against direct Special Power Damage but can be taken down with a bomb or multiple thrown objects.

If the Blimp's shield drops and it takes damage while the shield is lowered, the Blimp will divert into a defensive holding pattern while firing on the player.

| Variable | Explanation | |
|---|---|---|
| **Recovery laps** | The number of rotations around the player that the blimp must do to recover. | |
| **Movement Speed Increase Per Hit** | the amount added to the boss's total speed when ever it takes a hit. | |
| **Hits to Disable Shield** | The number of thrown objects it takes to disable the shield | |
| **Shield Heal Timer** | Removes 1 hit to the shield each countdown. | |
| **Shield Touch Damage** | How much damage the shield deals to the player. | |

**Player strategy**

1. Dodge incoming bombs by ducking and/or sidestepping.
2. Grab bombs off the ground and squeeze them, shutting the timer off and turning them into contact explosives.
3. Return the bomb to the sender by anticipating the bombing run and throwing the explosive as the Blimp's shield drops.

**And/Or:**

1. Attempt to disable the shield prematurely and land extra hits using rubble or the blimps own defensively deployed drones.

**And/Or:**

1. Attempt to short out the shield by baiting the Blimp into overheating it's guns while the player is using something as a shield.

The Blimps' shield redeploys and the boss attempts to resume it's Defensive Pattern.
The Blimp regenerates it's damaged weapon while inside the Shield and deploying drones. Once the weapons are regenerated it resumes the bombing run.

#### Staging

For each time the Blimp receives damage, the movement speed increases slightly, and it launches an additional bomb on its run and speeds up.

Once the player has successfully damaged the Blimp 3 times, explosions and particle effects go off the boss retreats into the horizon leaving a trail of smoke and sparks.

As with all player interaction, positive hit on Blimp should give the player clear feedback in the form of particle effects, sounds effects as well as animation (impulse from hit).

The boss will flash a darkened shade when damaged.

An undamaged Blimp emits only movement and attack particles.

A Blimp damaged once emits steam particles constantly as it leaks gas and occasional electrical arcs.

A Blimp damaged twice emits steam, electric arcs and is on fire.

| Variable | Description | |
|---|---|---|
| **Damage Speed Increase** | How many units are added to the blimps speed when it takes damage. Applied evenly across all move speeds (but not to accelleration). | |
| | | |

**Weapon configurations**

| Type | Ammo | Shots per burst | Pause between burst | Damage per shot |
|---|---|---|---|---|
| Gun Deck | Hitscan x2 per broadside | 1-4 | Distance Dependant | 20 |
| Larva Bomb | 1 + 1 per damage taken | NA | Pattern Dependant | TBD |

**Parameters (OLD)**

| Variable | Function | Type / Value(s) |
|---|---|---|
| AttackSpeed | Determines the ‘charging’ time of the boss' attack in seconds | Int |
| DownTime | Determines the length in seconds of the vulnerable period after attack. | Int |
| RetreatValue | The distance (in units) at which the boss will decide the player is ‘too close’ and retreat | Float |
| RetreatDistance | The distance in units which the boss will attempt to cover when retreating. | Float |
| RetreatMoveSpeed | The speed at which the boss traverses the RetreatDistance. | Float |
| RetreatDelay | The delay between when the Boss decides to retreat and when it | Int |

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| | executes the move in seconds. | |
| DefensiveCruisingSpeed | The speed at which the boss traverses its defensive holding pattern. | Float |
| TimeToRegenShields | The time in seconds that it takes for the shield to regenerate once downed. | Float |
| IdealAttackDistance | The distance in units from which the boss attempts to attack. | Float |
| MaxAttackDistance | The upper limit of the distance at which a boss can attack. | Float |
| AttackPower | The damage in HP that the boss' attack will inflict. | Int |
| AttackMoveSpeed | How fast the boss moves on its bombing run. | Float |
| TimeToLockAttack | The time interval in seconds after which an attack becomes unbreakable. | Int |
| ViableAttackDistance | The distance over which projectile attacks will still penetrate buildings. | Float |
| NormalMoveSpeed | The regular speed at which the boss moves | Float |

| | | |
|---|---|---|
| | when not retreating or attacking. | |
| BossHealth | The number of ‘hits’ the boss takes to disable. | Int |
| DamageType | The type of damage this enemy deals. | Bullet, Laser, Etc. |
| GraceTime | The time interval in seconds for which the boss waits before attacking. | Int |
| AggroDistance | The detection distance in units for the payer. | Float |
| InitialBombsToDrop | The number of bombs to drop on a bombing run. | Int |
| RevengeBombMultiplier | The number of bombs to add to the run for each point of damage the boss absorbs | Int |
| TimeToOverheatGundeck | The amount of sustained fire (in seconds) it takes to overheat the gundeck | TBD |
| | | |